

Joosung (Benjamin) Park

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EDUCATION

Penn State University — BS in Computer Science; Minors in Mathematics and Art Expected May 2027

Goal: learn each culture, find the value to contribute: prioritize learning and applying machine learning

EXPERIENCE

MC REU 2026 — Measuring Whether an AI Can Understand a Minecraft Builder's Intentions June – August 2026
from Raw Observation

- Built an end-to-end data pipeline — instrumented game capture, automated post-session processing, and human-in-the-loop labeling
- Trained a transformer decoder for next-token action prediction, reaching 71% top-1 accuracy on held-out real sessions (chance 1.2%) and 59% eight tokens ahead
- Repurposed frozen foundation models (VPT, MineCLIP, Uni3D) from action generation to perception to infer human intent, training only lightweight heads on project data

Penn State College of Engineering, University Park, PA

Undergraduate Research Ambassador (CEOI) June 2025 – Present

- Connect students with faculty, programs, and research resources across Penn State campuses to support undergraduate research engagement
- Advise peers on professional development strategies, sharing best practices for writing effective resumes and succeeding in interviews

Front Desk Student Worker August 2025 – Present

- Manage and organize in-person and hybrid student-advisor appointments, ensuring smooth and efficient check-ins

Academic Integrity Student Committee July 2026 – Present

- Serve as a student voice on academic integrity cases and policy within the College of Engineering

MC REU 2025 — Fine-Tuning LLMs for Code Smell Detection in C++ Game Development June – August 2025

- Fine-tuned two DeepSeek-Coder 1.3B models to detect code smells and generate automated fixes for open-source C++ game engine code
- Reduced GPU memory requirements with LoRA/QLoRA parameter-efficient fine-tuning, accelerating training iteration
- Developed a Python pipeline to collect and preprocess C++ repositories from GitHub into task-specific detection and fix-generation datasets

Honor Guard, Staff Sergeant, Republic of Korea Air Force August 2022 – June 2024

- Performed intricate drill routines with 100% precision at 100+ public events, national ceremonies, and state functions

PROJECTS

PathFinder — Yale Hackathon 2026 March 2026

- Engineered the AI/ML and backend stack of a satellite-image disaster-response tool in a two-person team: YOLO damage detection, FastAPI routing engine, and geo-referencing pipeline

RippleJourney — ripple-journey.vercel.app March 2026

- Built a full-stack app in 3 days (Next.js, TypeScript, Prisma, PostgreSQL with pgvector) that turns a raw research idea into a step-by-step roadmap with AI-matched faculty mentors via the Google Scholar API

LEADERSHIP & ACTIVITIES

Team Member, Builder's Collective — Penn State University: build with student founders and engineers at collegiate hackathons, most recently shipping PathFinder at Yale Hackathon 2026 2026 – Present

Member, Rowing Team (Crew) — Penn State University: train at daily 5:30 AM practices, learning the discipline and synchronized trust of crew while maintaining strong academics 2025 – Present

Representative, International Student Organization: mentored 10+ new international students, bridging Korean and American campus cultures through academic and cultural adaptation 2021 – 2022

Team Member, League of Legends Esports: analyzed opponents and set strategy with a five-player roster, winning the ECAC tournament 2021 – 2022